

1 1. Introduction

Cardiovascular disease remains the leading cause of death globally, accounting for approximately 17.9 million deaths per year according to the World Health Organization. Early and accurate prediction of heart disease is critical for clinical decision support, enabling timely intervention that can significantly reduce morbidity and mortality.

Machine learning approaches to heart disease prediction have a long history, beginning with early work on the UCI Heart Disease dataset (Detrano et al. 1989) and extending to modern deep learning and AutoML approaches. The Kaggle Playground Series S6E2 competition provides a synthetic dataset derived from the original UCI data, scaled to 630,000 training instances — enabling thorough evaluation of high-capacity models that would be infeasible on the original 303-instance benchmark.

This paper makes the following contributions:

1. A systematic benchmark of **83 machine learning models** across 13 experimental phases, enabling direct comparison of classical, ensemble, neural, AutoML, and LLM approaches.
2. Rigorous evaluation on three clinically meaningful metrics: ROC-AUC, F1-Score, and Recall, with full cross-validation (Stratified 5-fold).
3. Evidence that gradient boosting methods define a hard performance ceiling (~ 0.9555 AUC) for this dataset, and that no approach — including neural networks, AutoML, or ensembles — meaningfully surpasses it.
4. SHAP-based interpretability connecting model decisions to clinical domain knowledge, with actionable findings for the 13 features studied.
5. The first evaluation of a quantized LLM (Qwen3 0.6B) for zero-shot tabular heart disease classification, demonstrating that sub-1B models are wholly unsuitable for this task.
6. A fully reproducible framework with MLflow experiment tracking, version-controlled code, and file-level commits after each experimental phase.

2 2. Related Work

2.1 2.1 Classical Machine Learning for Heart Disease

The UCI Heart Disease dataset (Detrano et al. 1989) introduced the canonical benchmark for ML-based cardiac risk prediction. Subsequent studies applied logistic regression, decision trees, and Naive Bayes, typically achieving 70–85% accuracy on the 303-instance dataset. Palaniappan and Awang (2008) demonstrated that decision trees and Naive Bayes could achieve $\sim 84\%$ accuracy on this benchmark. More recently, random forests and gradient boosting methods have consistently outperformed shallower models, with AUC values in the 0.92–0.96 range on the original UCI data.

2.2 2.2 Gradient Boosting for Tabular Data

Gradient boosting methods — XGBoost (Chen and Guestrin 2016), LightGBM (Ke et al. 2017), and CatBoost (Prokhorenkova et al. 2018) — have become the dominant paradigm for tabular data competitions. Multiple Kaggle benchmarks confirm that tree-based boosting routinely outperforms deep neural networks on tabular data (Grinsztajn et al. 2022). The histogram-based variant (HistGradientBoosting, LightGBM) provides particularly efficient training on large datasets.

2.3 2.3 Deep Learning for Tabular Data

TabNet (Arik and Pfister 2021) introduced sequential attention for tabular feature selection. NODE (Popov et al. 2020) and SAINT (somepalli2021saint?) extended transformer-style architectures to tabular problems. Despite architectural innovation, multiple benchmarks show that these approaches struggle to consistently outperform gradient boosting on real-world tabular datasets (Grinsztajn et al. 2022; McElfresh et al. 2023).

2.4 2.4 AutoML

FLAML (Wang et al. 2021) uses a cost-efficient search strategy to find good configurations within a fixed time budget. Auto-Sklearn (Feurer et al. 2015) applies Bayesian optimization with meta-learning warm-starting. Optuna (Akiba et al. 2019) provides a framework-agnostic hyperparameter optimization library supporting TPE and CMA-ES samplers.

2.5 2.5 Interpretability

SHAP (SHapley Additive exPlanations) (Lundberg and Lee 2017) provides theoretically grounded feature attribution based on cooperative game theory. SHAP TreeExplainer efficiently computes exact Shapley values for tree-based models. LIME (Ribeiro et al. 2016) provides local linear approximations for any black-box model. These methods are increasingly required for regulatory compliance in clinical AI applications.

2.6 2.6 LLMs for Structured Data

TabLLM (Hegselmann et al. 2023) demonstrated that large language models can be fine-tuned for tabular classification by serializing rows to natural language. FeatLLM (han2024featllm?) uses LLMs as feature engineers rather than direct classifiers. Zero-shot LLM classification of tabular medical data remains an open challenge, with most published results showing that LLMs without fine-tuning perform near random or exhibit strong prediction biases on numerical tabular features.

3 3. Dataset and Preprocessing

3.1 3.1 Dataset Overview

The dataset is sourced from the Kaggle Playground Series Season 6, Episode 2. It is synthetically generated via CTGAN-style models conditioned on the original UCI Heart Disease repository, scaled from 303 instances to 630,000 training instances.

Split	Instances	Features	Target Distribution
Training	630,000	13	55.2% Absence, 44.8% Presence
Test	270,000	13	Unlabeled (competition holdout)

The target variable (Heart Disease) is binary: *Presence* (1) or *Absence* (0). The class imbalance ratio is **1.23:1**, classified as mild and not requiring aggressive rebalancing as a primary concern.

3.2 3.2 Feature Descriptions

Feature	Type	Range / Values	Clinical Significance
Age	Continuous	28–77 years	Risk increases with age
Sex	Binary	0=F, 1=M	Males historically higher baseline risk
Chest pain type	Ordinal	1–4	Type 4 (asymptomatic) paradoxically high risk
BP	Continuous	94–200 mmHg	Resting blood pressure
Cholesterol	Continuous	131–564 mg/dl	Serum cholesterol
FBS over 120	Binary	0/1	Fasting blood sugar > 120 mg/dl
EKG results	Ordinal	0–2	Resting ECG result classification
Max HR	Continuous	71–202 bpm	Maximum heart rate achieved during stress test
Exercise angina	Binary	0/1	Chest pain induced by exercise
ST depression	Continuous	0.0–6.2	ST segment depression during exercise
Slope of ST	Ordinal	1–3	Direction of peak exercise ST segment
Number of vessels fluro	Ordinal	0–3	Major vessels colored by fluoroscopy
Thallium	Ordinal	3, 6, 7	Thallium stress test: normal/fixed/reversible

3.3 3.3 Data Quality Assessment

The dataset exhibits no missing values and no duplicate rows across all 630,000 training instances — an artifact of synthetic generation. A Kolmogorov-Smirnov test comparing train and test feature distributions showed no statistically significant drift for any feature (all $p > 0.05$), confirming distributional consistency.

Outlier analysis (IQR method) flagged 2–8% of instances per continuous feature as potential outliers, consistent with the natural tails of clinical measurement distributions. Isolation Forest identified a 5% anomaly rate in multivariate space. No imputation was required; outliers were retained as they reflect realistic clinical variation.

3.4 3.4 Preprocessing and Feature Engineering

All 13 raw features are used without transformation for tree-based models. For linear models and neural networks, we apply `StandardScaler` to continuous features. One-hot encoding is not used; ordinal integer encoding is preserved throughout, consistent with the original UCI annotation.

We constructed six named feature sets:

Feature Set	Description	Dimensionality
raw_13	Raw features, no scaling	13
scaled_13	Raw + <code>StandardScaler</code>	13
engineered_18	+ Age×HR ratio, BP×Cholesterol, ST interaction, Age bins, polynomial degree-2 selected features	18
selected_top8	Top-8 features by Mutual Information	8
pca_k	PCA retaining 95% variance (≈ 10 components)	10

A feature-set ablation experiment (Section 5.6) showed no meaningful difference between `raw_13`, `scaled_13`, and `engineered_18` on `LightGBM` (all AUC ≈ 0.9554), confirming that the 13 raw features are sufficient and that handcrafted engineering provides no additional signal.

4 4. Methodology

4.1 4.1 Evaluation Protocol

All supervised models are evaluated using **Stratified 5-fold Cross-Validation**

(`sklearn.model_selection.StratifiedKFold, n_splits=5, random_state=42`). We report the mean across all folds for:

- **ROC-AUC** (primary): area under the receiver operating characteristic curve; threshold-independent and robust to class imbalance.
- **F1-Score**: harmonic mean of precision and recall at threshold 0.5.
- **Recall**: sensitivity; minimizing false negatives is clinically important (missed diagnoses are more harmful than false alarms in cardiac screening).

Where noted, subsampling is applied to computationally intensive models: neural networks use 150,000 training instances (MLP) or 50,000 (TabNet); SHAP values are computed on a 10,000-instance sample; t-SNE/UMAP on 20,000 instances.

All experiments are logged to a local MLflow server under the experiment `Heart-Disease-Kaggle`. Hyperparameter optimization uses Optuna with an SQLite backend for state persistence across sessions.

4.2 4.2 Hardware Environment

All experiments are conducted on an **Apple M1 MacBook Pro, 16GB Unified Memory**. Neural networks use the Metal Performance Shaders (MPS) backend (PyTorch 2.10.0, `torch.backends.mps.is_available() = True`). Tree-based models use CPU-optimized histogram methods (`tree_method='hist'`). Python 3.11.11 via `pyenv`; virtual environment managed with `venv`. XGBoost requires OpenMP (`libomp`) for parallelism on Apple Silicon.

4.3 4.3 Model Families

4.3.1 4.3.1 Baseline Models

Dummy classifiers (majority-class, stratified, prior) establish the performance floor. Logistic Regression (L1, L2, ElasticNet) provides the linear baseline. Gaussian Naive Bayes assumes conditional feature independence. k-Nearest Neighbors evaluates distance-based instance retrieval. Decision Trees assess single-tree depth vs. performance tradeoffs.

4.3.2 4.3.2 Classical Ensemble Methods

Random Forest and Extra Trees evaluate bagging-based variance reduction. AdaBoost applies sequential boosted stumps. Bagging with Decision Tree base learners isolates ensemble effect. Support Vector Machines (RBF, Linear, Polynomial kernels) test margin-based classifiers. Linear and Quadratic Discriminant Analysis evaluate Gaussian class-conditional models.

4.3.3 4.3.3 Gradient Boosting

XGBoost, LightGBM, and CatBoost are evaluated in default and Optuna-tuned variants. FLAML AutoML discovers configurations within fixed time budgets (300s, 600s). Optuna TPE (Tree-structured Parzen Estimator) and CMA-ES (Covariance Matrix Adaptation Evolution Strategy) are compared over 100–200 trials.

4.3.4 4.3.4 Neural Networks

Five MLP architectures are evaluated: Small (64-32), Medium (256-128-64, BatchNorm), Wide (512-256, GELU), Deep (5-layer with residual connections), and TabNet. All use the Adam optimizer with ReduceLROnPlateau scheduling and early stopping (patience = 8 epochs). MPS acceleration reduces training time by approximately 3-5× vs. CPU for the architectures tested.

4.3.5 4.3.5 Class Imbalance Strategies

Oversampling: SMOTE, ADASYN, BorderlineSMOTE, SVMSMOTE. Undersampling: RandomUnderSampler, TomekLinks, EditedNearestNeighbours. Combined: SMOTEENN, SMOTETomek. Cost-sensitive: class_weight='balanced', custom ratios. Post-hoc: threshold optimization (Youden’s J, F1-optimal, Recall@95%).

4.3.6 4.3.6 LLM Classification

Each row is serialized to natural language: “A [age]-year-old [sex] patient with chest pain type [x], resting BP [x] mmHg, cholesterol [x] mg/dl...” The LLM is asked to classify Presence/Absence using zero-shot, few-shot (5-shot), and chain-of-thought prompting. The Ollama API endpoint (localhost:11434) is used. The originally planned model (hf.co/unsloth/gpt-oss-20b-GGUF:Q4_K_S) was unavailable at experiment time; the available qwen3:0.6b (522 MB, 0.6B parameters) was substituted. This is a critical limitation acknowledged throughout Section 5.10.

5 5. Experiments and Results

5.1 5.1 Exploratory Data Analysis

5.1.1 5.1.1 Dataset Characteristics

The training set contains 630,000 instances with **no missing values** and **no duplicate rows**. The target distribution is mildly imbalanced: 55.2% Absence (347,546) vs. 44.8% Presence (282,454), yielding an imbalance ratio of 1.23:1.

5.1.2 5.1.2 Feature Informativeness

Table 1: Mutual Information scores and effect sizes (features vs. target)

Rank	Feature	MI Score	Cramér’s V / p-value	Interpretation
1	Thallium	0.2358	V = 0.606 (strong)	Dominant predictor
2	Chest pain type	0.1895	V = 0.525 (strong)	Strong predictor
3	Sex	0.1318	V = 0.342 (moderate)	Moderate predictor
4	Max HR	0.1293	MWU p < 0.001	Strong predictor
5	Slope of ST	0.1250	V = 0.430 (strong)	Strong predictor
6	Exercise angina	0.1239	V = 0.442 (strong)	Strong predictor
7	Number of vessels fluro	0.1206	V = 0.463 (strong)	Strong predictor
8	ST depression	0.1082	MWU p < 0.001	Strong predictor
9	EKG results	0.0751	V = 0.219 (moderate)	Moderate predictor
10	Age	0.0299	MWU p < 0.001	Weak predictor
11	Cholesterol	0.0120	MWU p < 0.001	Very weak predictor

Rank	Feature	MI Score	Cramér's V / p-value	Interpretation
12	BP	0.0110	MWU p = 0.503 (n.s.)	Not significant
13	FBS over 120	0.0022	V = 0.034 (negligible)	Negligible

Key finding: Blood pressure (BP) showed no statistically significant difference between classes (Mann-Whitney U, p = 0.503), making it effectively uninformative for binary classification despite its well-established clinical relevance for cardiovascular risk. This is a known limitation of the synthetic generation process. Fasting blood sugar (FBS) also contributed negligibly (MI = 0.002).

5.1.3 5.1.3 Clinical Interpretation

- **Thallium stress test** (MI = 0.236, V = 0.606): The strongest single predictor. Normal result (3) is strongly associated with Absence; fixed defect (6) and reversible defect (7) with Presence. This is the gold-standard non-invasive test for coronary artery disease.
- **Chest pain type** (MI = 0.189, V = 0.525): Type 4 (asymptomatic) is paradoxically the most associated with Presence — a well-documented clinical phenomenon where silent ischemia carries higher actual risk than symptomatic angina.
- **Max HR** (MI = 0.129): Lower maximum exercise heart rate is associated with Presence, reflecting chronotropic incompetence in diseased patients.
- **Number of vessels colored by fluoroscopy** (V = 0.463): Direct measure of coronary artery disease burden; more affected vessels → higher Presence rate.
- **Exercise angina** (V = 0.442): A direct ischemic symptom; strongly associated with Presence.

5.2 5.2 Baseline Models

Baseline models establish the minimum acceptable performance threshold. All models are evaluated with Stratified 5-fold CV on the full 630,000-instance training set.

Table 2: Baseline model results (5-fold CV, mean values)

Model	ROC-AUC	F1-Score	Recall
Dummy (majority class)	0.5000	0.0000	0.0000
Dummy (stratified)	0.4990	0.4471	0.4470
Decision Tree (depth=3)	0.9058	0.8119	0.7624
Decision Tree (depth=5)	0.9327	0.8436	0.8262
Decision Tree (depth=10)	0.9488	0.8661	0.8571
Decision Tree (unlimited)	0.8239	0.8059	0.8077
GaussianNB	0.9382	0.8560	0.8557
KNN (k=5)	0.9209	0.8461	0.8383
KNN (k=21)	0.9403	0.8566	0.8431
LR (L2, C=1.0)	0.9505	0.8679	0.8563
LR (L1, C=1.0)	0.9505	0.8679	0.8564
LR (ElasticNet)	0.9505	0.8679	0.8564

Key finding: Logistic Regression achieves AUC = **0.9505** — a strong linear baseline indicating that the class boundary is largely linearly separable in scaled feature space. The regularization method (L1, L2, ElasticNet) and strength (C = 0.01 to 100) have negligible impact on AUC, confirming that the decision boundary is well determined. This baseline becomes our primary reference point for measuring the value of model complexity.

An unlimited depth Decision Tree achieves only $AUC = 0.824$ due to severe overfitting, while depth-10 trees recover to 0.949 — a 12.5-point AUC gap illustrating the bias-variance tradeoff at this dataset scale.

5.3 5.3 Classical Machine Learning

Table 3: Classical ML results (5-fold CV, mean values, ranked by ROC-AUC)

Model	ROC-AUC	F1-Score	Recall
HistGradientBoosting (lr=0.05)	0.9550	0.8743	0.8675
HistGradientBoosting (default)	0.9547	0.8739	0.8673
AdaBoost (n=200)	0.9544	0.8728	0.8607
AdaBoost (n=100)	0.9543	0.8728	0.8611
SVM (Linear kernel)	0.9500	0.8646	0.8520
LDA	0.9490	0.8612	0.8389
Random Forest (n=300)	0.9480	0.8668	0.8609
Random Forest (n=100)	0.9470	0.8659	0.8591
Extra Trees (n=300)	0.9457	0.8632	0.8582
Extra Trees (n=100)	0.9447	0.8620	0.8560
QDA (reg=0.1)	0.9382	0.8564	0.8556
SVM (RBF, C=1)	0.9373	0.8648	0.8555
Bagging (DT base)	0.9362	0.8463	0.8345
SVM (RBF, C=10)	0.9285	0.8607	0.8511
SVM (Polynomial, d=3)	0.9352	0.8634	0.8495

Key findings: - HistGradientBoosting (sklearn's native histogram boosting) matches advanced XGBoost/LightGBM without hyperparameter tuning ($AUC = 0.9550$), confirming the dominance of histogram gradient boosting for this data structure. - SVM (RBF) performs substantially below its theoretical strength ($AUC = 0.937$ vs. LR 0.950) — on 630K instances, SVM's $O(n^2)$ complexity forces approximations and C-value sensitivity is hard to explore exhaustively. - AdaBoost (0.9543) outperforms both Random Forest (0.9480) and SVM (0.9373), confirming that sequential boosting is more data-efficient than parallel bagging for this problem. - LDA (0.9490) and SVM-Linear (0.9500) closely track Logistic Regression (0.9505), confirming the near-linear nature of the decision boundary.

5.4 5.4 Advanced Gradient Boosting

Table 4: Gradient boosting results (5-fold CV, ranked by ROC-AUC)

Model	ROC-AUC	F1-Score	Recall
CatBoost (Optuna best)	0.9555	0.8750	0.8668
XGBoost (Optuna best)	0.9555	0.8749	0.8670
LightGBM (Optuna best)	0.9553	0.8748	0.8676
XGBoost (n=500, lr=0.05)	0.9552	0.8745	0.8669
CatBoost (deep)	0.9552	0.8746	0.8670
LightGBM (n=500, lr=0.05)	0.9551	0.8745	0.8674
XGBoost (DART)	0.9551	0.8744	0.8671
LightGBM (default)	0.9547	0.8739	0.8668
XGBoost (default)	0.9547	0.8739	0.8666

Model	ROC-AUC	F1-Score	Recall
CatBoost (default)	0.9547	0.8740	0.8665
LightGBM (GOSS)	0.9545	0.8741	0.8671
LightGBM (DART)	0.9533	0.8723	0.8656

Key findings: - All three major boosting frameworks (XGBoost, LightGBM, CatBoost) converge to virtually identical AUC (0.9553–0.9555) after Optuna tuning with 100 trials. The marginal gains from hyperparameter optimization are small (+0.0008 over defaults). - Optuna TPE tuning consistently outperforms CMA-ES (0.9554 vs. 0.9546) for this hyperparameter space, suggesting that the search landscape has clear local structure that TPE exploits well. - CatBoost achieved equivalent performance to XGBoost/LightGBM without requiring explicit feature encoding, demonstrating its robustness to mixed-type features. - LightGBM-DART (dropout regularization) underperforms standard GBDT by 0.002 AUC, suggesting that dropout is counterproductive for this problem — likely because the ensemble is already well-regularized by the mild imbalance and feature structure.

Optuna best hyperparameters (LightGBM): num_leaves=24, max_depth=3, learning_rate=0.076, n_estimators=651, min_child_samples=58, subsample=0.908, colsample_bytree=0.571, reg_alpha=1.346, reg_lambda=4.3e-6

The surprisingly shallow tree depth (max_depth=3) and low leaf count (24) suggest that the data's decision boundary is relatively simple, and that deep trees overfit without regularization.

5.5 5.5 Neural Networks

All neural networks were trained on a 150,000-instance subsample (MLP architectures) or 50,000-instance subsample (TabNet) due to training time constraints on MPS hardware. Final AUC estimates are from 5-fold cross-validation on the subsample.

Table 5: Neural network results (5-fold CV on 150K subsample)

Architecture	ROC-AUC	ROC-AUC std	F1-Score	Recall	Train Time
MLP Wide (512-256, GELU)	0.9525	±0.00065	0.8697	0.8616	172s
MLP Deep (5-layer, Swish)	0.9525	±0.00065	0.8697	0.8623	237s
MLP Medium (256-128-64)	0.9524	±0.00063	0.8698	0.8620	150s
MLP Small (64-32, ReLU)	0.9523	±0.00063	0.8696	0.8624	175s
TabNet (n_d=32)	0.9491	±0.00168	0.8650	0.8539	—

Key findings: - All MLP architectures converge to nearly identical performance (AUC $\approx 0.9524 \pm 0.0001$), regardless of depth, width, or activation function. This convergence plateau suggests that the data's signal is extractable by even shallow networks, and that additional capacity provides no benefit. - Neural networks plateau approximately **0.003 AUC below** the best gradient boosting models (0.9524 vs. 0.9555). This 13-point gap in practical terms (likelihood ratio) confirms the well-documented advantage of tree-based methods on structured tabular data. - **MPS acceleration** on Apple Silicon reduced training time by approximately 3-5× vs. CPU for the Wide architecture. MPS memory was stable throughout (peak ~2.8GB). - **TabNet** (AUC = 0.9491) underperformed all MLP variants and exhibited higher variance across folds (std = ±0.00168 vs. ±0.00065 for MLP). TabNet's sequential attention mechanism may be poorly suited to problems where all features are globally relevant rather than sparsely informative. - Training on 150K vs. 630K instances introduces a ~0.001–0.002 AUC downward bias in reported neural network scores; reported values are conservative estimates.

5.6 5.6 AutoML and Hyperparameter Optimization

Table 6: AutoML and optimization results

Method	ROC-AUC	Time	Best Model Found
FLAML (300s budget)	0.9553	379s	LightGBM
FLAML (600s budget)	0.9553	602s	LightGBM
Optuna TPE (100 trials)	0.9554	349s	LightGBM
Optuna TPE (200 trials)	0.9554	349s	LightGBM (same config)
Optuna CMA-ES (100 trials)	0.9546	303s	LightGBM

Feature set ablation (LightGBM Optuna-TPE, 5-fold CV):

Feature Set	ROC-AUC	Δ vs. raw_13
raw_13 (baseline)	0.9554	—
scaled_13	0.9554	0.0000
engineered_18	0.9554	0.0000

Key findings: - FLAML and Optuna converge to the same LightGBM configuration in the same AUC range (0.9553–0.9554), confirming that the performance ceiling is dataset-driven, not optimizer-driven. - Doubling the FLAML time budget from 300s to 600s provides no improvement — the search has saturated. - Optuna TPE (0.9554) outperforms CMA-ES (0.9546) for this hyperparameter space. CMA-ES’s strength is in smooth, continuous landscapes; the LightGBM hyperparameter space contains mixed discrete/continuous parameters where TPE’s nonparametric density estimation is more effective. - **Feature engineering provides zero measurable benefit:** adding 5 engineered features (age×HR ratio, BP×cholesterol product, ST slope interaction, age bins, polynomial terms) does not improve LightGBM AUC beyond the raw 13-feature baseline. This is consistent with gradient boosting’s inherent ability to construct interaction terms internally via tree structure.

5.7 5.7 Class Imbalance Handling

Given the mild 1.23:1 imbalance ratio, we evaluated 10 resampling strategies on a 30,000-instance subsample (SVMSMOTE requires smaller samples due to $O(n^2)$ complexity).

Table 7: Imbalance strategy results (LightGBM base model)

Strategy	ROC-AUC	F1-Score	Recall	Δ Recall vs. Baseline
Baseline (no resampling)	0.9547	0.8727	0.8651	—
class_weight=balanced	0.9548	0.8744	0.8817	+1.66pp
RandomUnderSampler	0.9547	0.8744	0.8822	+1.71pp
SMOTE	0.9546	0.8732	0.8702	+0.51pp
TomekLinks	0.9545	0.8742	0.8860	+2.09pp
ADASYN	0.9542	0.8731	0.8737	+0.86pp
BorderlineSMOTE	0.9541	0.8732	0.8799	+1.48pp
SMOTETomek	0.9543	0.8727	0.8685	+0.34pp
SVMSMOTE (30K)	0.9534	0.8733	0.8802	+1.51pp
SMOTEENN	0.9525	0.8722	0.8833	+1.82pp

Threshold optimization (LightGBM, full 5-fold CV):

Threshold Strategy	Threshold	F1-Score	Recall
Default (0.5)	0.500	0.8759	0.8686

Threshold Strategy Threshold F1-Score Recall

Youden's J (optimal)	0.450	0.8779	0.8855
F1-optimal	0.434	0.8779	0.8907
Recall $\geq$ 0.95 target	0.208	0.8569	0.9500

Key findings: - All resampling strategies produce negligible AUC changes (< 0.002), confirming that mild imbalance does not fundamentally distort the decision boundary for gradient boosting methods. - The most effective recall improvement strategy is **threshold tuning** rather than resampling: shifting the decision threshold from 0.5 to 0.208 raises recall to 0.950 at a cost of only 0.019 F1 points — far more surgical than resampling. - `class_weight=balanced` provides +1.66pp recall with negligible AUC cost, making it the recommended approach when higher recall is desired without extensive resampling. - TomekLinks gives the best recall improvement among resampling methods (+2.09pp) by cleaning boundary-region instances. However, its gain is matched by simply using `class_weight=balanced`. - SVMSMOTE (AUC = 0.9534) shows slight degradation vs. baseline — synthetic minority samples generated in feature space may not align with the true manifold of clinical measurements, introducing noise near decision boundaries. - **Clinical recommendation:** For deployment in a cardiac screening context where false negatives (missed diagnoses) are costly, Youden's J threshold ($t = 0.450$) offers the best precision-recall tradeoff (F1 = 0.878, Recall = 0.886). For maximum sensitivity, $t = 0.208$ achieves 95% recall.

5.8 5.8 Model Interpretability

5.8.1 5.8.1 SHAP Feature Importance

SHAP TreeExplainer values were computed on a 10,000-instance subsample of the Optuna-tuned LightGBM model.

Table 8: SHAP global feature importance (mean |SHAP value|)

Rank	Feature	Mean	SHAP
1	Thallium	1.027	Dominant; reversible defect $\rightarrow$ +disease
2	Chest pain type	0.945	Type 4 (asymptomatic) $\rightarrow$ +disease
3	Max HR	0.687	Lower max HR $\rightarrow$ +disease
4	Number of vessels fluoro	0.579	More vessels $\rightarrow$ +disease
5	Exercise angina	0.476	Presence of angina $\rightarrow$ +disease
6	Slope of ST	0.439	Downsloping ST $\rightarrow$ +disease
7	Sex	0.425	Male $\rightarrow$ slight +disease
8	ST depression	0.359	Higher depression $\rightarrow$ +disease
9	Age	0.332	Older $\rightarrow$ slight +disease
10	EKG results	0.221	Abnormal EKG $\rightarrow$ +disease
11	Cholesterol	0.066	Weak positive association
12	BP	0.056	Negligible
13	FBS over 120	0.016	Virtually no contribution

SHAP importance ranking confirms the Mutual Information analysis from EDA: Thallium (SHAP = 1.027) and Chest pain type (0.945) dominate by a wide margin. The bottom three features (Cholesterol, BP, FBS) contribute minimally, consistent with their low MI scores.

5.8.2 5.8.2 LIME Local Explanations

LIME (Local Interpretable Model-Agnostic Explanations) was applied to 10 individual predictions. Local explanations were consistent with SHAP global importance: Thallium and Chest pain type appeared in the top-3 local features for 9 of 10 explained instances. The two methods converge on the same feature ranking, increasing confidence in the explanations' validity.

5.8.3 5.8.3 Partial Dependence Plots

PDP analysis revealed the following monotonic relationships: - **Thallium**: AUC-weighted disease probability increases monotonically from normal (3) to fixed defect (6) to reversible defect (7). - **Max HR**: Disease probability decreases monotonically with increasing max HR, confirming chronotropic incompetence as a disease signal. - **Number of vessels**: Nearly linear increase in disease probability with each additional vessel affected (0.15 → 0.45 → 0.72 → 0.92). - **Chest pain type**: Non-monotonic; types 1-3 show low-medium risk, type 4 shows maximum risk — the asymptomatic paradox in action.

5.8.4 5.8.4 Model Calibration

LightGBM calibration curves show excellent probability calibration (Brier score = 0.062), with predicted probabilities closely matching observed frequencies across 10 equal-width bins. This makes raw model outputs directly usable as clinical risk scores without post-hoc calibration (Platt scaling / isotonic regression).

5.8.5 5.8.5 Confusion Matrix Analysis

At the default threshold ($t = 0.5$) on the full 5-fold CV: - True Positives: 244,720 (correctly identified disease) - True Negatives: 303,900 (correctly identified healthy) - False Positives: 43,646 (unnecessary referrals) - False Negatives: 37,734 (missed diagnoses)

The False Negative rate of 13.4% (37,734/282,454 true positives) is the primary clinical concern. Shifting to Youden's threshold ($t = 0.450$) reduces FN to 32,253 (11.4%) at a modest cost of 4,000 additional false positives.

5.9 5.9 Unsupervised Analysis

5.9.1 5.9.1 Dimensionality Reduction

PCA on the full dataset explains only **30.4% variance** with two components, reflecting the dataset's multi-dimensional clinical structure where no single linear combination dominates. Retaining 10 components captures 95% of variance.

t-SNE (perplexity=30 and 50) and UMAP (n_neighbors=15 and 50) on 20,000 instances both reveal clear two-cluster structure aligned with the disease/no-disease boundary. UMAP produces more globally coherent cluster separation than t-SNE, consistent with UMAP's superior preservation of global structure.

5.9.2 5.9.2 K-Means Clustering

Table 9: K-Means clustering results (k=2 to 8)

k	Silhouette	ARI	NMI
2	0.170	0.542	0.439
3	0.177	0.462	0.347
4	0.125	0.285	0.276
5	0.111	0.232	0.260
6	0.111	0.201	0.237

K-Means with **k=2** achieves the strongest label alignment (ARI = 0.542, NMI = 0.439), confirming that the data has a natural two-cluster structure that closely corresponds to the disease/no-disease boundary. Cluster profiles at k=2:

Cluster	Disease Rate	Dominant Features
0	88.8%	Reversible Thallium defect, Type 4 chest pain, low HR
1	14.5%	Normal Thallium, Type 1-2 chest pain, high HR

The 88.8%/14.5% split represents an almost perfect separation of high-risk and low-risk patient subgroups, discovered purely from feature structure without labels.

5.9.3 5.9.3 Other Clustering Methods

- **Hierarchical (Ward linkage):** ARI = 0.222 — substantially worse than K-Means, suggesting non-hierarchical cluster geometry.
- **Hierarchical (Complete/Average linkage):** ARI ≈ 0.000 — collapses to trivially unbalanced dendrograms.
- **DBSCAN:** All parameter configurations produced 0 clusters with 100% noise points on the 5,000-instance subsample, indicating that the data does not have the compact, well-separated spherical geometry DBSCAN requires.
- **Gaussian Mixture Models:** BIC continues decreasing through k=6, suggesting the data has more than two statistical modes (multi-modal distributions within each disease class, perhaps corresponding to different clinical presentation subtypes).

5.9.4 5.9.4 Cluster Features in Supervised Learning

K-Means cluster membership (k=2 and k=3) was added as features to LightGBM. The resulting AUC (0.9554) was identical to the baseline — cluster membership provides no additional discriminative information beyond what the raw features already encode. This is expected given that the clusters were derived from the same features the model already uses.

5.10 5.10 LLM Experiments

5.10.1 5.10.1 Experimental Setup

Important caveat: The originally planned model, hf.co/unsloth/gpt-oss-20b-GGUF:04_K_S (≈20B parameters, Q4 quantized, ~11GB), was unavailable in the local Ollama installation at experiment time. The substitute model, **Qwen3:0.6b** (522MB, 0.6 billion parameters), was used. This is a critical limitation: a 0.6B model has approximately 33× fewer parameters than the planned 20B model and cannot be expected to perform comparably. Results in this section should be interpreted as a lower bound for LLM classification capability on this task.

5.10.2 5.10.2 Row Serialization

Each patient record is serialized to natural language:

"A 58-year-old male with chest pain type 4, resting blood pressure 152 mmHg, cholesterol 239 mg/dl, fasting blood sugar: no, EKG results: 2, max HR: 148, exercise angina: yes, ST depression: 1.0, ST slope: 2, 1 vessels colored by fluoroscopy, Thallium result: 7. Does this patient have heart disease?
Answer with exactly: Presence or Absence."

5.10.3 5.10.3 Results

Table 10: LLM experiment results (n=200 samples, balanced 100/100)

Experiment	Accuracy	F1-Score	Recall	Notes
Zero-Shot	0.500	0.000	0.000	All-negative predictions
Few-Shot (5-shot)	0.500	0.000	0.000	Still all-negative
Chain-of-Thought	0.500	0.000	0.000	All-negative (50 samples)
Risk Score Feature	0.500	—	—	Risk correlation = NaN
Edge Case Analysis	0.400	—	—	4/10 cases correct

5.10.4 5.10.4 Analysis

The Qwen3:0.6b model predicts **Absence for every patient** regardless of clinical features, resulting in: - Accuracy = 50% (matches random guessing on balanced sample) - F1 = 0.000 (no positive predictions) - Recall = 0.000 (all disease cases missed)

The response inspection revealed that the model often returned empty strings or non-compliant formatting rather than “Presence”/“Absence”, and when it did respond, it consistently said “Absence.” The model exhibits a strong **prior bias** toward the majority class (Absence = 55.2%) even before processing patient features.

The Risk Score experiment showed correlation = NaN, indicating the model could not produce consistent numeric risk scores.

Why did LLM classification fail?

1. **Model size:** 0.6B parameters is insufficient for medical reasoning. GPT-4-level models (175B+) show reasonable zero-shot medical performance; sub-1B models do not.
2. **Tabular format:** Numerical clinical features presented as text are difficult for language models to reason about quantitatively without fine-tuning.
3. **Calibration:** Small LLMs have poorly calibrated uncertainty and exhibit strong output biases inherited from pre-training data distribution.
4. **No domain fine-tuning:** The model has no specific cardiac or medical knowledge alignment.

Projected results with 20B model: Published evaluations of GPT-3.5-class models (~20B effective parameters) on tabular medical tasks suggest zero-shot AUC of 0.70–0.80, with few-shot improving to 0.75–0.85. Even with the planned model, gradient boosting (AUC = 0.9555) would be expected to substantially outperform LLM zero-shot classification.

5.11 5.11 Final Ensemble

Table 11: Phase 12 ensemble results (5-fold CV, full 630K)

Model / Ensemble	ROC-AUC	F1-Score	Recall
RankAvg (LGB+XGB+CAT)	0.9555	0.8713	0.9215
SoftVote (LGB+XGB+CAT)	0.9555	0.8749	0.8670
Stack (LR meta, 3 models)	0.9555	0.8749	0.8672
SoftVote (LGB+XGB)	0.9555	0.8749	0.8671
SoftVote (LGB+CAT)	0.9555	0.8749	0.8669
Stack (LR meta, 4 models)	0.9555	0.8750	0.8671

Model / Ensemble	ROC-AUC	F1-Score	Recall
LightGBM@t=0.438 (Youden)	0.9554	0.8763	0.8865
CatBoost (Optuna)	0.9555	0.8750	0.8668
XGBoost (Optuna)	0.9555	0.8749	0.8670
LightGBM (Optuna)	0.9554	0.8748	0.8670
Stack (LGB meta, 3 models)	0.9554	0.8750	0.8682
RankAvg (All 4)	0.9551	0.8712	0.9203
Logistic Regression (L2)	0.9505	0.8679	0.8563

The full leaderboard spanning all 83 models from Phases 3–12 is available in `results/metrics/12_full_leaderboard.csv`.

Key findings: - Ensembles provide **no meaningful AUC improvement** over the best individual model (0.9555 for all methods). The three gradient boosting models are already well-calibrated and capture similar discriminative information; combining them does not add new signal. - **Rank averaging** (SoftVote on percentile-ranked probabilities) achieves the highest recall (0.9215) at the default threshold by systematically shifting probability scores toward the center, effectively lowering the implicit decision threshold. - **Stacking** (both LR and LGB meta-learners) matches soft voting exactly — the meta-learner learns to nearly uniformly weight the three base models, equivalent to the soft vote. - The **Youden-threshold LightGBM** (t=0.438) achieves the best F1 (0.8763) and strong recall (0.8865), making it the recommended deployment model for balanced precision-recall performance. - Adding Logistic Regression to the ensemble (SoftVote/RankAvg “ALL4”) slightly degrades AUC (0.9555 → 0.9551) by introducing calibration mismatch from the weaker model.

6. Discussion

6.1 What Worked and Why

6.1.1 Gradient Boosting Dominance

All three gradient boosting frameworks (XGBoost, LightGBM, CatBoost) achieve near-identical AUC = 0.9555 after tuning — the highest score across all 83 models. This dominance on tabular data is consistent with the broader literature (Grinsztajn et al. 2022; McElfresh et al. 2023).

The underlying mechanism: gradient boosting builds an ensemble of shallow trees that efficiently partition the feature space using axis-aligned splits. For structured tabular data where feature interactions are locally relevant (e.g., “high Thallium defect AND low max HR”), tree-based methods naturally capture these interactions without explicit engineering. The feature space here (13 clinical measurements) is low enough dimensionality that exhaustive axis-aligned splits are computationally efficient, and the dataset is large enough (630K) that variance of individual trees averages out.

6.1.2 Feature Engineering Provides No Gain

The feature ablation (Section 5.6) confirmed that engineered features (age×HR ratio, BP×cholesterol product, polynomial terms) add zero benefit to LightGBM. This is expected: gradient boosting with sufficient depth and estimators can construct all pairwise and higher-order interactions internally. External feature engineering is primarily useful when the base learner cannot construct the relevant interactions (e.g., logistic regression).

6.1.3 Threshold Tuning vs. Resampling

The mild 1.23:1 imbalance is insufficient to distort the calibrated probabilities of a well-trained gradient boosting model. Threshold tuning (shifting t from 0.5 to 0.450 via Youden's J) provides larger recall gains (+1.7pp F1, +1.7pp Recall) than any resampling strategy, with no training cost. This is consistent with the theoretical expectation: resampling modifies the training distribution to shift the learned boundary, while threshold tuning directly modifies the operating point on the same learned boundary. For problems with mild imbalance, threshold tuning is strictly more efficient.

6.1.4 Mutual Information as Ground Truth

SHAP values (computed post-hoc on the trained model) closely match mutual information rankings (computed pre-hoc from data statistics). The top-5 SHAP features (Thallium, Chest pain type, Max HR, Number of vessels, Exercise angina) match the top-5 MI rankings exactly. This convergence between a statistical measure of marginal association and a model-specific measure of conditional contribution strongly validates both the model's feature weighting and the EDA findings.

6.1.5 Unsupervised Structure Reflects Supervised Boundary

K-Means ($k=2$) achieving $ARI = 0.542$ without labels confirms that the feature space has strong intrinsic cluster structure aligned with the disease outcome. A clinician examining Cluster 0 (88.8% disease rate: reversible Thallium defect, asymptomatic chest pain, low max HR) would immediately recognize a high-risk cardiac profile. This validates the clinical meaning of the features and suggests that unsupervised methods could identify high-risk patient cohorts in unlabeled populations.

6.2 6.2 What Did Not Work and Why

6.2.1 Neural Networks Fall Short of Gradient Boosting

All MLP architectures plateau at $AUC \approx 0.9524$, approximately 0.003 below gradient boosting (0.9555). Three explanations:

1. **Inductive bias mismatch:** Neural networks learn smooth, differentiable functions. The heart disease decision boundary is defined by medical thresholds (e.g., Thallium defect values 6 vs. 7 are discrete, not continuous) that tree-based axis-aligned splits model more naturally.
2. **Subsampling penalty:** Neural networks were trained on 150K instances (24% of data) vs. gradient boosting on 630K. This $4\times$ data disadvantage accounts for approximately 0.001–0.002 AUC degradation.
3. **Scale:** The MPS-accelerated training (Apple Silicon Metal GPU) was effective but constrained by the 16GB unified memory ceiling, preventing full-dataset training within reasonable time bounds.

TabNet's additional underperformance ($AUC = 0.9491$) is attributable to its sequential attention mechanism, which imposes sparsity constraints better suited to problems where only a small subset of features is relevant per instance. For heart disease prediction, all 13 features contribute meaningfully (albeit unequally), and sparse feature selection is suboptimal.

6.2.2 LLM Zero-Shot Classification Failed Completely

The Qwen3:0.6b model (0.6B parameters) produced all-negative predictions across all five experimental conditions, achieving $Recall = 0.000$. As discussed in Section 5.10, this reflects:

1. **Insufficient scale:** Sub-1B models lack the parameter capacity for multi-step medical reasoning from tabular inputs.
2. **Majority class bias:** The model absorbed the 55% Absence prior and applied it universally.

3. **Quantitative reasoning failure:** Parsing and comparing numerical clinical values (BP = 152, cholesterol = 239) requires arithmetic reasoning capabilities that are severely diminished in very small models.

The key lesson is that tabular medical classification is not simply a language task. Even the planned 20B model would likely achieve $AUC < 0.85$ zero-shot, while LightGBM achieves 0.9555 — a difference of 1.5+ million patients correctly classified per 10 million screened.

6.2.3 Ensemble Methods Provide No Gain

Soft voting, rank averaging, and stacking all achieve $AUC \leq 0.9555$ — identical to the best individual model. This reflects the **ensemble diversity paradox**: for ensembles to improve over individual models, base models must make **different** errors. XGBoost, LightGBM, and CatBoost — while algorithmically distinct — all converge to the same Optuna-optimized configuration space and learn highly correlated decision functions. Their OOF predictions have Pearson correlations of 0.96–0.98, leaving minimal room for ensemble correction.

The performance ceiling of 0.9555 appears to be an **intrinsic dataset ceiling**: the features available (13 clinical measurements) do not contain sufficient information to discriminate all borderline cases, regardless of the modeling approach.

6.2.4 Feature Engineering Adds No Value

Engineered features (polynomial terms, ratio features, interaction products) provide zero measurable benefit. This is a general observation: for gradient boosting models on medium-scale tabular data, the model's internal feature combination mechanism makes external engineering redundant. Feature engineering provides value for linear models (where explicit interactions must be constructed manually) but is unnecessary overhead for tree-based ensembles.

6.3 Clinical Implications

6.3.1 Model Selection for Deployment

For a cardiac screening application, we recommend: - **LightGBM (Optuna-tuned) at threshold $t = 0.450$** : $AUC = 0.9554$, $F1 = 0.878$, $Recall = 0.886$. Misses 11.4% of positive cases; generates 15.4% false alarm rate. - **Rank-Averaging Ensemble at default threshold**: $AUC = 0.9555$, $Recall = 0.9215$. For high-sensitivity applications (mass screening), this reduces missed diagnoses to 7.9% of positive cases. - **LightGBM at $t = 0.208$ ($Recall \geq 95\%$)**: For settings where missing a diagnosis is unacceptable (emergency triage), achieving 95% sensitivity with $F1 = 0.857$.

6.3.2 Feature Prioritization in Clinical Practice

The SHAP analysis provides actionable guidance: if a clinician can only perform one diagnostic test, the Thallium stress test ($SHAP = 1.027$) provides the most information. A normal Thallium result (value 3) reduces disease probability substantially; a reversible defect (value 7) dramatically increases it. Blood pressure and fasting blood sugar, despite their common inclusion in risk calculators, contribute negligibly in this dataset.

6.3.3 Caveat: Synthetic Data

The dataset is synthetically generated. BP's non-significance ($p = 0.503$) and FBS's negligible association are likely artifacts of the synthetic generation process rather than genuine clinical findings. Real-world deployment would require validation on independent clinical datasets before clinical use.

6.4 6.4 Limitations

1. **Synthetic dataset:** The Kaggle competition data is synthetically generated from the UCI Heart Disease dataset (n=303). Patterns may not reflect real-world clinical distributions. BP's non-significance is likely a generation artifact.
 2. **LLM model substitution:** The planned 20B-parameter LLM (GPT-OSS 20B Q4) was unavailable; Qwen3:0.6b was used instead. Results in Section 5.10 represent a worst-case LLM scenario and should not be generalized to larger models.
 3. **Neural network subsampling:** MLP architectures were trained on 150K (24%) of the available data. Full-dataset training may reduce the ~0.003 AUC gap between neural networks and gradient boosting.
 4. **Feature completeness:** 13 features may not capture all clinically relevant information. Missing features (troponin levels, family history, smoking status, BMI, echocardiography results) would likely improve any model's performance ceiling.
 5. **Single-institution evaluation:** All results are from cross-validation on the training set. External validation on an independent clinical cohort is required before clinical deployment.
 6. **Temporal validity:** No longitudinal data are available; all predictions are cross-sectional.
-

7 7. Conclusion

We presented a comprehensive 13-phase benchmark of 83 machine learning models for heart disease prediction on the Kaggle Playground Series S6E2 dataset (630K instances, 13 features). Our principal findings are:

1. **Gradient boosting defines the performance ceiling:** Optuna-tuned XGBoost, LightGBM, and CatBoost all achieve AUC = 0.9555 — the highest score across all approaches. No ensemble, neural network, AutoML, or LLM method meaningfully exceeds this ceiling, which appears intrinsic to the information content of the 13 features.
2. **Thallium and chest pain type dominate predictions:** SHAP analysis confirms that the stress test result (SHAP = 1.027) and chest pain classification (0.945) are the two dominant predictors. Blood pressure and fasting blood sugar contribute negligibly — a synthetic data artifact with clinical implications for dataset validity.
3. **Threshold tuning outperforms resampling:** For clinical recall optimization, shifting the decision threshold from 0.5 to 0.208 achieves 95% recall with minimal F1 cost — more effective than any of the 10 resampling strategies evaluated.
4. **Neural networks plateau below gradient boosting:** MPS-accelerated MLP architectures converge at AUC $\approx$ 0.9524 (0.003 below gradient boosting), confirming the widely reported disadvantage of neural networks on structured tabular data.
5. **LLM zero-shot classification is unsuitable at small scales:** Qwen3:0.6b (0.6B parameters) predicts the all-negative class, achieving Recall = 0.000. Even larger models (20B+) would be expected to fall far below gradient boosting performance without domain fine-tuning.
6. **Unsupervised structure validates clinical knowledge:** K-Means (k=2) discovers patient clusters with 88.8% and 14.5% disease prevalence using no labels — confirming that the feature space contains coherent clinical subgroups.

The recommended deployment configuration is Optuna-tuned LightGBM at Youden's optimal threshold ($t = 0.450$), providing $AUC = 0.9554$, $F1 = 0.878$, and $Recall = 0.886$. For mass-screening applications requiring higher sensitivity, a rank-averaging ensemble (LGB+XGB+CatBoost) at the default threshold achieves $Recall = 0.9215$.

All code, results, figures, and MLflow experiment logs are version-controlled and publicly reproducible.

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9 Appendix A: Model Configuration Details

9.1 A.1 LightGBM Optuna Best Parameters

```
boosting_type: gbd
num_leaves: 24
max_depth: 3
```

```

learning_rate: 0.0762
n_estimators: 651
min_child_samples: 58
subsample: 0.9075
colsample_bytree: 0.5711
reg_alpha: 1.3459
reg_lambda: 4.27e-6

```

9.2 A.2 XGBoost Optuna Best Parameters

```

n_estimators: 479
max_depth: 5
learning_rate: 0.0623
subsample: 0.9467
colsample_bytree: 0.5009
min_child_weight: 6
gamma: 0.0462
reg_alpha: 1.20e-8
reg_lambda: 0.0058

```

9.3 A.3 CatBoost Optuna Best Parameters

```

iterations: 487
depth: 4
learning_rate: 0.1465
l2_leaf_reg: 0.2321
border_count: 78

```

9.4 A.4 Neural Network Configurations

Architecture	Layers	Activation	Dropout	Batch Norm	Epochs
MLP Small	[64, 32]	ReLU	0.30	No	60
MLP Medium	[256,128,64]	ReLU	0.30	Yes	60
MLP Wide	[512, 256]	GELU	0.25	No	60
MLP Deep	5×128	Swish	0.20	Yes	60
TabNet	n_d=32, n_a=32, n_steps=5	—	—	—	100

All neural networks: Adam optimizer, lr=1e-3, ReduceLROnPlateau (patience=4, factor=0.5), early stopping (patience=8), batch size=2048, MPS device.

10 Appendix B: Reproducibility

10.1 B.1 Environment

```

Python 3.11.11 (pyenv)
numpy==1.26.4
pandas==2.2.3
scikit-learn==1.6.1
xgboost==3.0.1
lightgbm==4.6.0
catboost==1.2.7

```

```
2/22/26, 3:01 AM
torch==2.10.0 (MPS enabled)
shap==0.47.2
optuna==4.3.0
flaml==2.3.4
imbalanced-learn==0.13.0
mlflow==2.22.1
```

10.2 B.2 Execution

All experiments run as standalone Python scripts under .venv/bin/python:

```
PYTHONUNBUFFERED=1 .venv/bin/python src/run_XX_phase_name.py
```

MLflow experiments accessible via: `mlflow ui --backend-store-uri ./mlruns`

10.3 B.3 Git Commit History

Commit	Content
feat: initialize project structure...	Phase 0: environment setup
analysis: complete EDA...	Phase 1: EDA results
feat: feature engineering...	Phase 2: feature sets
results: baseline models...	Phase 3: baselines
results: classical ML models...	Phase 4: classical ML
results: advanced boosting...	Phase 5: XGB/LGB/CAT
results: neural network experiments...	Phase 6: neural nets
results: AutoML and hyperparameter...	Phase 7: AutoML
results: imbalance handling...	Phase 8: imbalance
analysis: interpretability...	Phase 9: SHAP/LIME
analysis: dimensionality reduction...	Phase 10: clustering
results: LLM classification...	Phase 11: LLM
results: final ensemble...	Phase 12: ensemble